

Lab 2 — Backpropagation by Hand, and CNNs

Machine Learning II · Tutorial

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Wednesday, October 14, 2026 · 19:00–22:00

Goal of the session. Finish the theory of Homework 1 (XOR with ReLU, Xavier/He, Adam's bias correction), then Homework 2 in full: convolution and pooling by hand, receptive fields, parameter counts, and the gradients through a convolutional layer — each verified against PyTorch.

Materials

- `homework_1_theory.pdf` — Problems 4–6.
- `homework_2_student.pdf` / `homework_2_solutions.pdf` — the 13 CNN problems.
- `homework_2_student.ipynb` / `homework_2_solution.ipynb`.
- Lecture 2 of ML II (optimizers, initialization, dropout, batch norm, convolution, receptive field).

1 Homework 1, Theory (50 min)

Task 1 — XOR with ReLU (Problem 4)

(a) Prove that no network without a hidden layer computes XOR: a single unit is a linear classifier and XOR is not linearly separable (ML I, Lecture 2: the perceptron mistake bound needs a margin, and here none exists). (b)–(d) Build the minimal ReLU network explicitly and verify all four inputs in numpy.

Task 2 — Xavier and He (Problem 5)

Derive $\text{Var}(z_k) = n_{\text{in}} \text{Var}(w) \text{Var}(x)$ for a linear unit with independent zero-mean weights and inputs, hence $\text{Var}(w) = 1/n_{\text{in}}$ (Xavier). Redo it for ReLU, where half the pre-activations are zeroed, to get $\text{Var}(w) = 2/n_{\text{in}}$ (He). Then verify in torch: initialize a 10-layer ReLU network both ways, push a batch of $\mathcal{N}(0,1)$ inputs through it, and plot the activation variance per layer.

Task 3 — Adam's bias correction (Problem 6)

Show $\mathbb{E}[\mathbf{m}_t] = (1 - \beta_1^t) \mathbf{g}$ under stationary gradients, hence the correction $\hat{\mathbf{m}}_t = \mathbf{m}_t / (1 - \beta_1^t)$. Compute the uncorrected bias at $t = 1, 10, 100$ for $\beta_1 = 0.9$, $\beta_2 = 0.999$. Then run Exercise 5

of the Homework 1 notebook (SGD, momentum, RMSprop, Adam on one problem) and relate the curves to the derivation.

2 Homework 2: Convolutions (70 min)

Task 4 — by hand, then by torch (Problems 1–6)

1. Parameter count: a fully connected layer from 28×28 to 28×28 vs. one 3×3 convolution with 16 filters. Compute both; then read them off `sum(p.numel() for p in layer.parameters())`.
2. 2-D convolution of a 5×5 input with a 3×3 kernel, by hand, stride 1, no padding. Check with `torch.nn.functional.conv2d`. Note that torch computes cross-correlation; flip the kernel to see the difference.
3. Output size $\lfloor (n + 2p - k)/s \rfloor + 1$: verify on four (n, k, s, p) combinations.
4. Receptive field: two 3×3 convolutions give RF 5; add a 2×2 max-pool (jump 2) and a third convolution; compute the RF and confirm by perturbing one input pixel and seeing which outputs change.

Task 5 — pooling and architectures (Problems 7–10)

Max vs. average pooling on a hand-made 4×4 map. LeNet-5's parameter count, layer by layer, matched against the torch model. Then Problem 10: show that two stacked 3×3 convolutions have the receptive field of one 5×5 with $18c^2$ instead of $25c^2$ parameters, and one more nonlinearity.

Task 6 — backpropagation through a convolution (Problems 11–13)

Derive $\partial L / \partial \mathbf{K}$ for a single-channel convolution: it is the cross-correlation of the input with the upstream gradient. Implement it in numpy and compare with `autograd` on a random input. Then the two pooling gradients: max-pool routes the gradient to the argmax (sparse); average-pool spreads it uniformly (dense). Verify both.

3 Exercises (to submit before Lab 3)

Exercise 1. The bonus section of the Homework 2 notebook: an end-to-end MNIST CNN. Report test accuracy after 3 epochs and the number of parameters; compare with the ≤ 100 -unit MLP from Lab 1's exercise.

Exercise 2. Homework 1 notebook, Exercise 6: the dying-ReLU experiment and dropout vs. ℓ_2 . Connect the ℓ_2 result to ridge regression (ML I, Lecture 3): what plays the role of λ ?

Submission

Notebook `lab02_lastname_firstname.ipynb` and a PDF of the derivations, before Wednesday, October 21, 18:00.